### 1. Indicator 9: Understand the relationship between the context of the problem and the representation of the mathematical solution. Use this understanding to interpret the solution in context and gauge its feasibility and possible limitations

- Highly relevant.
  Students must grasp the meaning of the percentage statement:

$$\text{TV-owning households} = 85.8\% \times \text{Total households}$$

Therefore, total households = $\frac{2.8\text{ million}}{0.858} \approx 3.26\text{ million}$ , which is closest to option C (3.3 million).
This process requires linking the mathematical operation (division) to its real-world implication—namely, that the total number of households must be greater than the number owning TVs—and assessing whether the computed value is plausible.

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### 2. Indicator 1: Interpret a mathematical result back into the real-world context

- Highly relevant.
  The calculated value (~3.26 million) is not an abstract number; it must be understood as "the total number of households in Switzerland." Students need to recognize that this figure represents actual social units within a national population and match it to the appropriate order of magnitude among the answer choices.

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### 3. Indicator 2: Identify whether a mathematical result or conclusion does, or does not, make sense given the context of the problem

- Highly relevant.
  Students can quickly eliminate implausible options using contextual reasoning:
  - Option A (2.4 million) is less than 2.8 million (the number of TV-owning households), which is impossible;
  - Option B (2.9 million) implies over 96% of households own TVs, contradicting the given 85.8%.
    Such commonsense checks based on the problem context are central to solving this item.

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### 4. Indicator 5: Interpret mathematical outcomes presented in various formats in relation to a situation; compare or evaluate two or more representations in relation to a situation

- Moderately relevant.
  The data are presented in a table format, combining an absolute number (2.8 million) and a relative proportion (85.8%). Students must translate between these representations: from "part / whole = percentage" to "whole = part ÷ percentage." This involves integrating different forms of quantitative information.

### 5. Indicator 6: Use knowledge of how the real world impacts the outcomes and calculations of a mathematical procedure or model to make contextual judgments about how results should be adjusted or applied

- Moderately relevant.
  Although the problem does not require explicit model adjustment, students implicitly apply real-world knowledge: the total number of households must exceed the number that own TVs. This commonsense constraint directly informs their evaluation of answer choices, illustrating how real-world logic shapes the interpretation of mathematical results.

### 1. 第9条：Understand the relationship between the context of the problem and representation of the mathematical solution. Use this understanding to help interpret the solution in context and gauge the feasibility and possible limitations of the solution

**（理解问题情境与数学解表示之间的关系，并据此解释结果、评估其可行性与局限性）**

- 高度相关。
  学生需理解“85.8% of all households own TVs”这一百分比模型的含义：

$$\text{TV-owning households} = 85.8\% \times \text{Total households}$$

因此，总户数 = $\frac{2.8\text{ million}}{0.858} \approx 3.26\text{ million}$，最接近选项 C（3.3 million）。
这一过程要求学生将数学运算（除法）与现实意义（总户数应大于拥有电视的户数）联系起来，并判断结果是否合理（如不能小于2.8 million）。

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### 2. 第1条：Interpret a mathematical result back into the real world context

**（将数学结果重新解释回现实世界情境）**

- 高度相关。
  计算出的数值（约326万）不是一个抽象答案，而必须被理解为“瑞士全国的家庭总数”。学生需意识到这个数字代表真实人口结构中的家庭单位，并与选项中的数量级进行匹配。

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### 3. 第2条：Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem

**（判断数学结果在问题情境下是否合理）**

- 高度相关。
  学生可快速排除不合理选项：
  - A（2.4 million）< 2.8 million（拥有电视的户数），显然不可能；
  - B（2.9 million）仅略高于2.8 million，意味着97%以上家庭有电视，但题干明确说只有85.8%，故也不合理。
    这种基于常识和上下文的合理性检验是本题的关键策略。

### 4. 第5条：Interpret mathematical outcomes in a variety of formats in relation to a situation; compare or evaluate two or more representations in relation to a situation

**（结合情境解读不同形式的数学结果；比较或评估多个表示）**

- 中等相关。
  题目以表格形式呈现数据，包含绝对数（2.8 million）和相对比例（85.8%）。学生需在两种表示之间转换：从“部分/整体 = 百分比”推导出“整体 = 部分 ÷ 百分比”。这涉及对不同数据表示的理解与整合。

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### 5. 第6条：Use knowledge of how the real world impacts the outcomes and calculations of a mathematical procedure or model in order to make contextual judgments about how the results should be adjusted or applied

**（利用现实世界知识判断数学结果应如何调整或应用）**

- 中等相关。
  虽然本题未要求调整模型，但学生隐含使用了现实常识：总户数必须大于拥有电视的户数。这种常识直接影响对选项的筛选，体现了现实约束对数学结果的修正作用。